



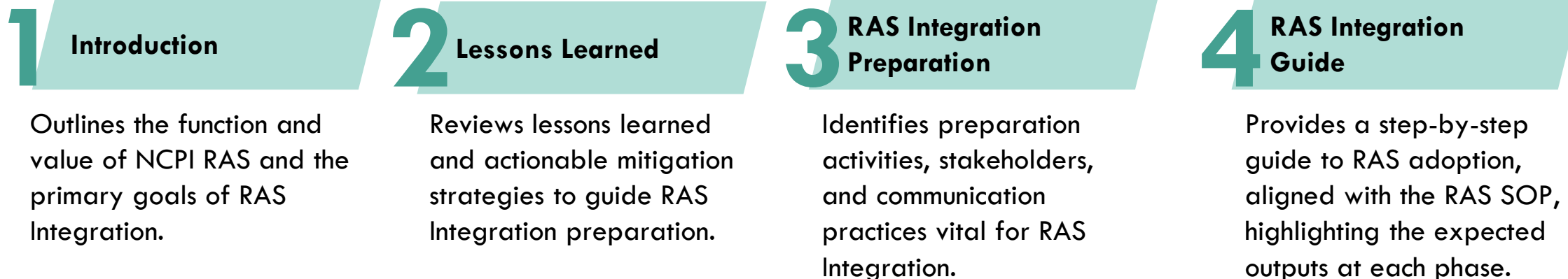
NIH Cloud Platform Interoperability (NCPI) Research Auth Service (RAS) Adoption Playbook



Overview: NCPI RAS Adoption Playbook

This document outlines how to plan and execute RAS integrations, highlighting past lessons learned, integration preparation, and providing a step-by-step playbook to reduce potential challenges in future RAS integrations.

Purpose:



To Promote:

A seamless, enjoyable NCPI Partner System RAS Integration experience with:

- A clear vision and plan for RAS Integration
- Tools to identify and mitigate potential challenges

...and enable long term results with:

- A federated data ecosystem enabling interoperability empowering the scientific community
- Uniform, secure, and user-friendly authentication and authorization technology



Introduction to NCPI

The [NIH Cloud Platform Interoperability \(NCPI\) Program](#), supported by The Office of Data Science Strategy (ODSS), establishes and implements guidelines and technical standards to empower end-user analyses across participating platforms and facilitates the realization of a trans-NIH, federated data ecosystem.

A federated data ecosystem endeavors to enable seamless search, access, and integration of datasets across participating systems and associated users/researchers.

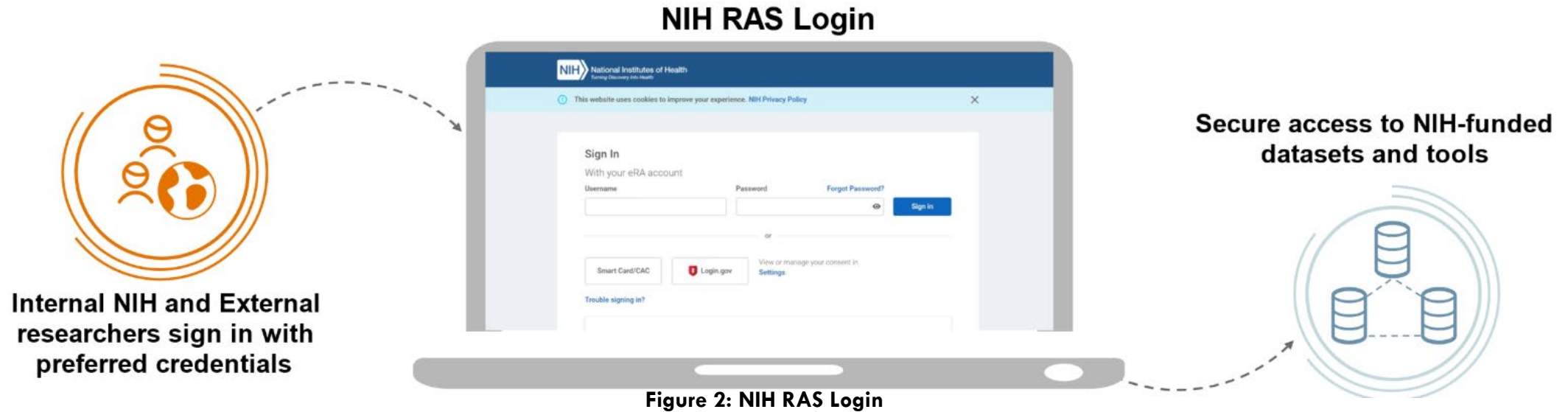
By focusing on interoperability, the NCPI program is ensuring that researchers can both find and integrate data more easily from the current participating NCPI Partners, as seen in Figure 1.



Figure 1: NCPI Participating Partners (2025)



Introduction to RAS for NCPI



The NCPI program is implementing **Researcher Auth Service (RAS)**, which supports a Single Sign-On (SSO) workflow. RAS is a service provided by NIH's Center for Information Technology (CIT) to facilitate access to NIH's open and controlled data assets and repositories, in a secure, consistent, and user-friendly manner. The NCPI RAS initiative aims to:

- Enhance NCPI Interoperability by centralizing authorization and authentication for NCPI Partners
- Enhance data security by replacing the legacy authorization mechanism of dbGaP Telemetry Files with RAS passports
- Strengthen NCPI Partner data governance and adopt modern security standards

Introduction to Authentication and Authorization

Authentication

Authentication refers to the process of verifying the identity of a user, process, or device, often as a prerequisite to allowing access to resources in an information system ([NIST](#)).



Application within RAS

A researcher's identity will be recognized regardless of which of their account credentials are entered ([RAS User Guide](#)).

Authorization

Authorization refers to the process of verifying the rights or permissions that are granted to a system entity to access a system resource ([NIST](#)).



Application within RAS

Researchers will be able to access data, through RAS Passports, according to the permissions that they have been granted by NIH, within each data repository. This enables stronger data security compliance while using the [GA4GH standard](#) ([RAS User Guide](#)).

Key RAS Integration Concepts:

[Identity Assurance Level \(IAL\)](#) refers to the strength or trustworthiness of digital identity information. In the NIH RAS context, systems set up their connections based on the level of identity proofing they require. IAL1 uses basic identity verification methods, while IAL2 involves more thorough checks with personal information.

[Identity Linking](#) RAS provides account identity consolidation so researchers can move from system to system using one set of credentials. Linking your identities mean you'll no longer need to remember multiple credentials to sign in.



Value of RAS for Scientific System Interoperability

The benefits of RAS for scientific systems and platforms include:



1. Enabling NCPI Interoperability

RAS

- **Centralized Authentication & Authorization:** RAS enables interoperability of NCPI Partner Systems and their scientific research platforms through centralized authentication and authorization. This allows RAS Partners to perform federated analysis across the NIH ecosystem, with access to over 40 PB of data. RAS also offers IAL1 and IAL2, an assurance approach which protects user's identities while balancing security and convenience.
- **RAS Passports:** The RAS Passport leverages the Global Alliance for Genomics and Health (GA4GH) specification to manage and verify researcher permissions for accessing NIH Controlled-Access Data (CAD). A RAS Clearinghouse verifies the RAS Passport being presented is a real RAS Passport by verifying that the issuer is NIH and that it has been signed by NIH RAS. Subject to refresh policies, the RAS passport confirms a user's permission to access NIH datasets, streamlining access and supporting seamless collaboration across NCPI partner platforms.
- **Application Programming Interfaces (APIs):** The RAS API Gateway provides a single, secure way to access NIH-funded data, whether it's stored at NIH or in external repositories. RAS's standardized API endpoints allow NCPI Partners to easily connect and tailor their systems, enabling different software platforms to exchange data seamlessly.
- **User Experience (UX):** RAS enhances UX by providing Single Sign-On (SSO) capabilities. During an active session, RAS automatically presents the necessary token when a user navigates to another NCPI Partner System page, granting access without the need for additional logins or authorization steps. This seamless process allows NIH researchers to search and access resources more efficiently.



Value of RAS for Data Security

The benefits of RAS for scientific systems and platforms include:



2. Enhancing Data Security

- **Identity Providers (IdPs) Brokering:** Brokering allows RAS to connect with multiple IdPs (Login.gov, eRA Commons, ID.me, InCommon Federation, and Google for Authentication only) to authenticate. Brokering different IdPs streamlines user access, strengthens security, reduces integration complexity, and enhances organizational agility.
- **Multi-Factor Authentication (MFA):** MFA is a security process that requires users to provide two or more forms of verification to confirm their identity before granting access to a system or application. RAS enforces MFA for IdPs, other than Google and eRA Commons, enhancing security by making it harder for attackers to gain access even if one credential (like a password) is compromised.
- **Identity Proofing:** Identity proofing is the process of verifying that a person is who they claim to be. RAS requires users to register with a trusted IdP at IAL2 or higher (requiring a higher level of identity proofing) for enhanced identity assurance and security when accessing NIH Controlled Access Data Repositories (CADRs). This significantly reduces the risk of unauthorized access due to compromised credentials.
- **Audit Logging:** RAS generates a transaction identifier to be used by RAS and clients to log and audit activities associated with RAS interactions. All activity against RAS endpoints are audited, recording user actions and system events to track access, changes, and potential security incidents.



RAS

Value of RAS for Compliance & Standards

To strengthen NCPI Partners data governance and adopt modern security standards, RAS:

- ✓ Adheres to the **Global Alliance for Genomics and Health (GA4GH)** standards for integration of researcher-focused applications and data repositories over the OIDC platform.
- ✓ Supports the **NIH Strategic Plan for Data Science** by advancing data infrastructure and ecosystem goals.
- ✓ Follows and adheres to **NIST (National Institute of Standards and Technology) 800-53 and 800-57** guidelines pertaining to configuration management, least privilege, and cryptographic key establishment & management.
- ✓ Applies **Office of Management and Budget (OMB)** direction on federal information policy and information technology.
- ✓ Enforces security measures and requirements in Partner System **Interconnection Security Agreements (ISAs)**.

RAS Integration Lessons Learned

*Review of RAS Integration Lessons Learned, Actionable Mitigation Strategies,
and RAS Integration Applicability*

Approaching RAS with Complete Requirements

- **Lesson Learned:** RAS has multiple service offerings and capabilities (authentication, authorization, & audit logging). Partner System teams should understand their systems complete requirements before approaching the RAS team, so the teams know what is possible for their RAS integration and what will be follow-on/future efforts.
- **Actionable Strategies:**
 - Meet with the RAS Team to understand the complete requirements for RAS integration prior to beginning any RAS integrations. This will allow the RAS Team to understand the Partner Systems needs and allows the Partner System to understand what is possible with RAS. If you begin the RAS SOP without complete requirements, it may lead to iterative rework and multiple review/approval cycles for RAS documentation, increasing overall integration timelines.
 - Confirming whether an ISA is required with the RAS Team, and initiating it early, is essential to avoid delays at the Development Phase Stage Gate. If needed, begin ISA process in Planning and be aware of associated timelines (net new ISAs typically require 2-6 months for approval while enhancements to existing ISAs typically take 2-4 months for approval).
 - Note: If a Partner System is integrating with RAS for both authentication and authorization, it is in the system's best interest to complete both integrations at the same time, rather than phasing them separately. However, if a Partner System chooses to phase its RAS integration, (integrating authentication first, followed by authorization at a later stage) it is important to understand that each RAS integration phase will require a full RAS SOP, including documentation, testing, and rollout.
 - **Applicability:** All phases of RAS Integration



Resilience Planning for RAS Integration Timelines

- **Lesson Learned:** By implementing resilience planning, such as schedule buffering into RAS integration timelines, teams are better equipped to accommodate evolving NIH data security requirements, adapt to new or modified RAS features, manage potential delays in documentation production and review cycles, and address unforeseen technical or funding challenges.
- **Actionable Strategies:**
 - Conduct an introductory meeting with the RAS Team before the Planning Phase to identify Partner System data security requirements and desired RAS features, clarifying the anticipated [level of effort](#) and enabling the inclusion of adequate schedule buffers in the integration timeline. Continue to engage with the RAS Team as needed throughout all project phases to ensure timelines remain realistic and aligned.
 - Estimate and build schedule buffers into project timelines to account for Partner System documentation development and RAS review time. Identify and address all discoverable issues during documentation development, to the fullest extent possible, in order to minimize prolonged documentation review cycles and reduce the likelihood of timeline delays.
 - Incorporate schedule buffers into project timelines to mitigate potential delays caused by funding or staffing gaps, given that funding decisions are often outside the direct control of Partner Systems.
 - In the event of unexpected obstacles or changes, provide timely and transparent updates to RAS and all stakeholders involved in the project, ensuring alignment on any revised target dates for RAS implementation
- **Applicability:** All phases of RAS Integration



Facilitating Effective Communication

- **Lessons Learned:** Effective and regular communication among all key stakeholders – including Partner System Teams, RAS Team, Partner System Platform Teams, System Owner, Office of Data Science Strategy (ODSS), and users – is essential for achieving a successful RAS Integration.
- **Actionable Strategies:**
 - To ensure timely and targeted communication, utilize the suggested [RAS integration effective communication practices](#):
 - Schedule and manage bi-weekly syncs (Tiger Teams) with Partner Systems, currently managed by the NCPI ACC, to foster collaboration, address and discuss integration challenges and issues, and create RAS SOP documentation. Bring the RAS Team into Tiger Teams, as needed, for discussion and direct issue resolution.
 - Exchange regular email communications on documentation progress to keep all stakeholders and timelines informed.
 - Monitor RAS Team response to GitHub tickets and NCPI ACC/Partner System Team activity in Slack channels to facilitate real-time discussion and feedback.
 - Maintain weekly status trackers, ensuring visibility and alignment across multiple stakeholders.
 - To streamline communication and address recurring challenges, leverage NCPI ACC's standardized [Tips and Tricks Templates](#) to document processes effectively and proactively resolve common roadblocks.
- **Applicability:** All phases of RAS Integration



Proactive ISA Development and Renewal

- **Lesson Learned:** An important stage gate within the RAS Integration process is the development or renewal of the Information Sharing Agreement (ISA). The ISA process, for both development and renewal, is lengthy, sometimes taking months to receive necessary updates and signatures. The ISA must be signed and in place at the end of the Testing Phase, before Partner Systems can proceed to RAS Deployment.
- **Actionable Strategies:**
 - System Owners and Partner System Teams are advised to proactively discuss with the RAS Team if ISA development or renewal is required for their integration during the Planning Phase. If needed, begin ISA process in Planning and be aware of associated timelines (net new ISAs typically require 2-6 months for approval while enhancements to existing ISAs typically take 2-4 months for approval).
 - System Owners are encouraged to engage IC security teams during ISA development or renewal to ensure alignment with Partner System information sharing policies and to benefit from their specialized expertise.
 - The NCPI ACC instituted several ISA check-points built into the RAS Integration Playbook starting at the [Planning Phase](#) to ensure timely completion of the ISA prior to RAS Deployment.
- **Applicability:** Planning, Design, Development, and Testing Phases



Accommodating New Requirements for RAS Integration

- **Lesson Learned:** In the case that a partner system's requirements for RAS integrations necessitate net new requirements or features (e.g., adopting a new identity provider (IdP) or enabling RAS passport repackaging between subsystems, etc.), it is essential to engage the RAS Team early to gauge level of effort and allocate sufficient time and resources for development.
- **Actionable Strategies:**
 - Partner System Teams must work with the RAS Team at the earliest stages of the integration process to understand any known net new requirements for RAS Integration and assess potential impacts on integration timelines.
 - Following the determination of the level of effort for net new requirements in collaboration with the RAS Team, Partner System Teams should revise RAS integration timelines to accommodate any anticipated additional RAS development activities and timelines.
 - Integration timelines should be communicated to the Partner System Team, RAS Team, and NCPI ACC to ensure collective awareness and effective tracking.
- **Applicability:** All phases of RAS Integration



Utilizing RAS Integration Resources

- **Lesson Learned:** Leverage available RAS Integration resources, such as the [RAS Documentation for Partner Systems](#), [RAS Project Partner Development Guide](#), NCPI ACC [Tips and Tricks Templates](#), and the [RAS Service Offerings](#) page to simplify documentation preparation. These tools make the RAS integration process clearer and more manageable, especially for teams new to RAS, and help improve both the accuracy and speed of documentation review.
- **Actionable Strategies:**
 - Utilize [RAS Documentation for Partner Systems](#), [RAS Project Partner Development Guide](#), and the [RAS Service Offerings](#) page to understand RAS parameters, processes, and services.
 - Utilize the NCPI ACC's series of [Tips and Tricks Templates](#), with in-line guidance for completing documents required for the RAS integration process. Documents include: [IRE](#), [TDD](#), [Test Plan](#).
- **Applicability:** Planning, Design, Development, and Testing Phases



Preparing Users for RAS Deployment

- **Lesson Learned:** It is advantageous for Partner System Teams to proactively assess and plan for the downstream user impacts of new processes associated with RAS Deployment to ensure a seamless transition for all system users. Following the deployment of RAS Integration into production, users will experience changes to existing workflows. In certain instances, process updates may require users to create accounts for login through compatible Identity Providers (IdPs).
- **Actionable Strategies:**
 - Partner System Teams should develop and disseminate system-specific user guides prior to the live RAS integration. This will ensure that end users are adequately prepared to utilize RAS, understand its functionality, and are informed about the process for reporting issues following production deployment.
 - User workflows should be taken into consideration through RAS Planning and Design phases.
 - Users should also receive a communication notifying them of changes and next steps prior to deployment.
- **Applicability:** Deployment Phase



RAS Integration Preparation

Partner System Team preparation, identification of stakeholders, and communication best practices required for RAS Integration

Starting the RAS Integration Process



1. Read the RAS Integration Standard Operating Procedures (SOP) and Addendum to:

- Identify which key stakeholders are responsible for completing RAS documentation and who is accountable for document sign-off.
- Understand the required 'Stage Gate' activities within each phase of RAS integration that Partner Systems must complete before advancing to the next phase.
- Familiarize your Partner System and stakeholders with the communication practices between key stakeholders involved in RAS Integration, including the preferred channels for sharing updates, the frequency of meetings or status reports, protocols for escalating issues, and methods for documenting decisions and action items to ensure transparency and alignment throughout the integration process.



2. Understand Partner System requirements in the RAS Step-by-Step Playbook

- This step confirms that Partner Systems are fully prepared and have direct access to supplemental guidance, minimizing delays and maximizing the likelihood of a smooth integration.



3. Proceed with RAS Integration

- After completing the initial preparation steps, Partner Systems are ready to begin the RAS integration process and should proceed by emailing the RAS Project Team (NIHRASTeam@mail.nih.gov) about starting their integration.
- At this stage, Partner Systems can take advantage of several resources leveraging the NCPI ACC's [tips and tricks documentation](#), [templates](#), and historical integration knowledge to mitigate process issues and minimize delay cycles with the RAS Team.



Stakeholders of RAS Integration: What to Expect

Successful integration of RAS requires collaboration and coordination among key stakeholders across technical, operational, and scientific domains. Understanding who will be involved and what roles they play is essential for effective planning and execution.

Partner System Team

The **Institutes and Centers (IC) System Owners** are the key decision makers responsible for approval of key Partner System documentation. Additionally, they oversee data repositories, define integration requirements, ensure compliance with RAS standards, participate in integration meetings, and supervise their teams throughout the entire RAS integration lifecycle.

The **IC Partner System Team** integrates IT systems with NIH RAS services, gathers requirements, ensures security compliance, manages documentation, and collaborates with the IC Partner System Owner (SO), RAS Team, and NCPI ACC during RAS implementation.

RAS Team

RAS Leadership (ODSS/CIT) are the key decision makers responsible for team operations and the approval process of RAS documentation.

The **RAS Contractor Team** manages integration and user support, oversees partner systems and documentation, operates and phases access to RAS environments, and develops RAS software using Agile methodology.

NCPI ACC RAS Adoption Team*

The **NIH Cloud Platform Interoperability Administrative Coordinating Center (NCPI ACC)** oversees RAS adoption for NCPI Partner Systems, reviews NCPI Partner System documentation, guides IC Partner System Teams, and facilitates communication and issue resolution during RAS integration.



Communication Best Practices

Clear and timely communication is vital for stakeholder alignment and RAS integration success. This slide details recommended communication practices including what, how, and when to effectively communicate throughout RAS Integration.

	PURPOSE	CHANNEL	STAKEHOLDERS	EXAMPLE
	General Updates and Q&A	Email	Partner System Team, RAS Team	Partner System Team provides an update or asks a question on documentation to the RAS Team via email. Note: To ensure accurate tracking and status updates, copy all RAS Stakeholders in email communications.
	Documentation Submission	Email, Box	Partner System Team, NCPI ACC	Partner System Team notifies the RAS Team of documentation submission via email, confirming the estimated turnaround time for RAS review and including the document Box link. Note: if you don't have access to the RAS Box (or the links in this document), please ask the RAS team.
	Documentation Review & Approval	Email, Box	RAS Team	The RAS Team provides documentation feedback or sign-off and notification of progression to the next phase of RAS integration to Partner System Teams via email.
	Team-to-Team Engagement	Slack	Partner System Team, NCPI ACC	Partner System Teams have the choice to create Slack channels with sub-system/platform stakeholders and utilize it for informal updates and questions pertaining to RAS Integration.
	Reporting Issues and Troubleshooting	GitHub	Partner System Team, RAS Team, NCPI ACC	Partner System Teams utilize GitHub Tickets to notify the RAS Team of technical issues found in development and testing. The RAS Team tracks and addresses GitHub tickets by offering effective mitigation and resolution strategies.

Expected Level of Effort (LOE)

Partner Systems that are new to RAS integration (authentication and authorization) should plan for an LOE consistent with the timeline shown below. Partner Systems that have already implemented RAS authentication and are preparing for RAS authorization can generally expect a more expedited timeline.

Level of effort (LOE) — For systems starting RAS Integration (Authentication and Authorization)

Phase	Partner System Preparation	RAS Review	Total Estimated LOE for Phase
Planning	≈3 weeks	≈3 weeks	≈1.5 Months
Design	≈4-6 weeks	≈4-6 weeks	≈2-3 Months
Development	≈4-6 weeks	≈4-6 weeks	≈2-3 Months
Testing	≈4-6 weeks	≈4-6 weeks	≈2-3 Months
Deployment	≈1-2 weeks	≈1-2 weeks	≈0.5-1 Month

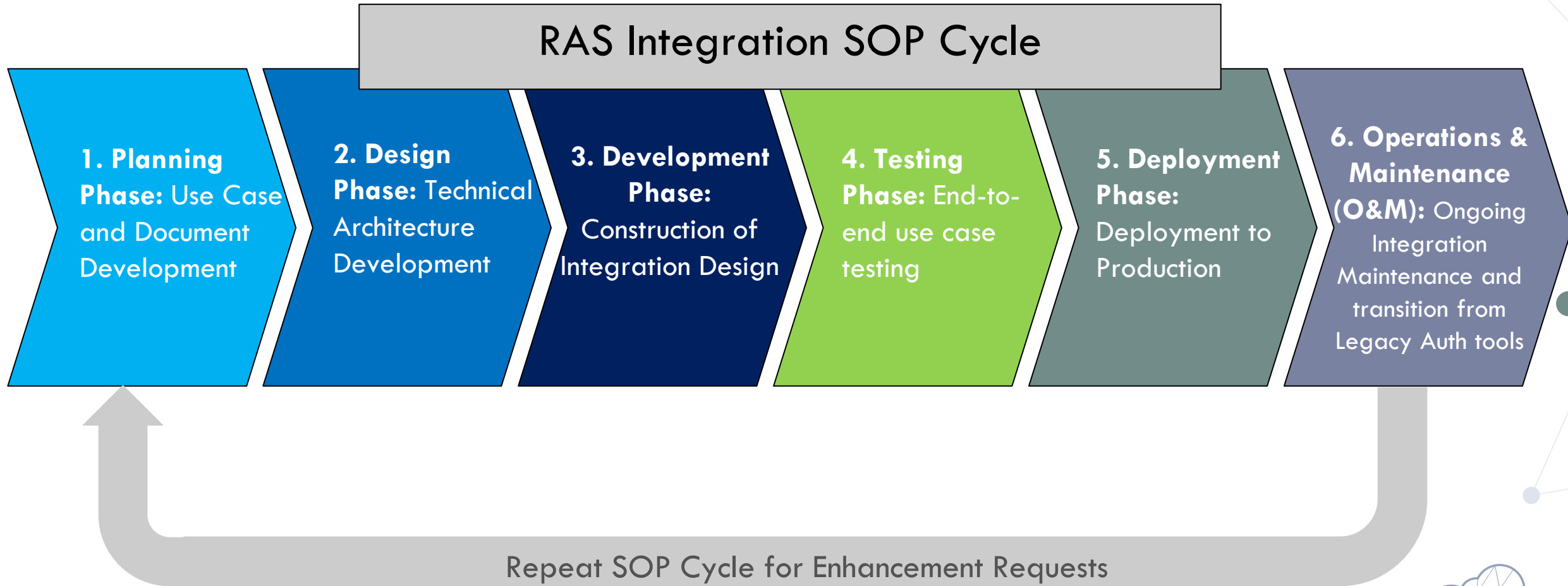


RAS Integration Guide

Summary, outputs, and step-by-step guide to the RAS Integration SOP Cycle

RAS Integration SOP Cycle: Summary Flow

The below diagram represents the phases of RAS Integration according to the SOP beginning with the Planning Phase.



Planning Phase Summary and Outputs

1. Planning

2. Design

3. Development

4. Testing

5. Deployment

6. O&M



Summary

The Partner System, RAS Team, and the NCPI ACC work together to understand the system requirements, share system information, and determine a preliminary strategy for the integration process. Creating a shared preliminary strategy during Planning is crucial for aligning expectations, proactively identifying risks, and estimating timeline requirements for a successful RAS integration.



Actions and Outputs

1. The Partner System Team completes initial meeting with the RAS Team to identify RAS integration goals, technical requirements, and timeline, specific to the Partner System.
2. The RAS Team creates and grants access to a [Box](#) folder which the Partner System Team will use to house key RAS integration documentation.
3. The Partner System Team will complete the [Integration Request Form \(IRF\)](#) and share a copy with the RAS Team for review and feedback. After finalization and with System Owner sign-off, the Partner System Team submits the IRF to the RAS Team for review and sign-off.
4. The Partner System Team begins the process of completing or updating the [ISA](#), which is a security agreement between the Partner System Team and the RAS Team.



Planning Phase Playbook

Step 1: Gather System Details for SOP:

Use the following questions as a framework to explore and document critical aspects of your partner system. While these questions are optional, your responses will provide valuable input for developing the necessary documents within the RAS SOP:

- **What is the primary purpose of your Partner System?**
 - Identify the core function your system serves within the organization.
- **Who are the key stakeholders?**
 - List main stakeholders (roles or groups).
- **What sub-systems or platforms make up the Partner System?**
 - Name each sub-system/platform (e.g., Gen3, PIC-SURE, Terra, Velsera) and summarize its primary use case (e.g., data analysis, data viewing, data mapping).
- **Where are the sub-systems/platforms hosted?**
 - Specify for each: on-premise, cloud, hybrid, or other environments.
- **Which user types access the sub-systems/platforms?**
 - (e.g., internal staff, external partners, researchers, international users, administrators)
- **Will the integration with RAS provide authentication, authorization, or both for each user type and use case?**
 - Specify per system area, if applicable.

Utilize the RAS
Playbook in tandem
with the RAS SOP



RAS SOP -
Planning (Pg. 12)



Planning Phase Playbook



Step 1: Gather System Details for SOP (Continued)

Use the following questions as a framework to explore and document critical aspects of your partner system. These questions are not mandatory, but they will help inform required documents within the RAS SOP:

- **How many users access the partner system each day?**
 - Provide estimates for user access across sub-systems/platforms
- **Which identity providers are currently supported for your users (including international users and projections for users with higher security needs)?**
 - List currently offered IdPs and indicate which are or are not supported by RAS.
- **What additional IdPs, if any, will need to be integrated for your Partner System?**
 - Specify gaps or anticipated needs
- **Based on data sensitivity and regulatory needs, what level of identity assurance is required (e.g., NIST IAL1 or IAL2)?**
 - Map assurance levels to user types and data types involved.
- **Are there sub-systems that require only RAS authentication versus those needing both authentication and authorization?**
 - Identify any exceptions within the overall Partner System.
- **Does your Partner System have requirements not currently supported by RAS?**
 - For example, additional IdPs, developer authentication, non-standard authorization models, integration with other services.
- **Are there Partner System-specific workflows, security concerns, audit/compliance needs, or technical constraints the RAS Team should understand to enable a secure and seamless integration?**



Planning Phase Playbook



Step 2: Assign Roles and Align on Integration Requirements

Meet jointly with the RAS Team to confirm key roles, clarify roles and responsibilities, and establish clear expectations for the RAS integration process. Build a shared understanding of all integration requirements including thorough review and discussion of system responses to the guiding questions in Step 1.

- **Outcome:** Agreement to proceed with RAS integration. The RAS Team will then:
 - Schedule the official onboarding kickoff meeting.
 - Provide Box folder access for reference and submission of integration documents.

Step 3: Draft the Integration Request Form (IRF)

The Partner System Team drafts the IRF, capturing all integration details, use cases, and user workflows.

- Utilize the NCPI ACC's [IRF tips and tricks template](#) to help with completeness and alignment with best practices.
- (Past Process) Partner Systems can share the IRF with the NCPI ACC for an informal RAS review to gain feedback based on prior RAS integrations and help align the IRF with RAS expectations.

Step 4: Determine Interconnection Security Agreement (ISA) Requirement

Discuss your Partner System's ISA requirements with the RAS Team. If needed, net new ISAs typically require 2-6 months for approval while enhancements to existing ISAs typically take 2-4 months for approval. Confirming whether an ISA is required with the RAS Team, and initiating it early, is essential to avoid delays at the Development Phase Stage Gate.

- Note: The ISA is the most frequent cause of delays in the RAS process due to the extensive review and signature process.



Planning Phase Playbook



RAS SOP -
Planning (Pg. 12)

Step 5: Submit IRF for RAS Review and Complete/Update ISA (as needed)

Submit the finalized IRF, signed off by the System Owner, to the designated [Box](#) folder.

- Email the RAS Team (NIHRASTeam@mail.nih.gov) following the [communication plan](#) guidelines with the document [Box](#) link, confirmation of System Owner approval and submission, and confirm the expected review turnaround time (typically 1–2 weeks).
- If designated necessary in Step 4, begin creating or updating the [ISA](#), ensuring all information reflects the current integration plans.

Step 6: Understand and Address RAS Review of IRF

When the RAS Team provides feedback, promptly address, acknowledge, and/or revise the IRF as needed, then resubmit to maintain RAS SOP momentum.

- If changes were made, ensure Box folder has the most recent document version and email off of original submission informing RAS that all IRF comments are addressed/closed out.
- If the RAS Team requests additional details for the Technical Design Document (TDD), ensure those points are captured and incorporated during the [Design](#) phase.

Step 7: Move to the Design Phase

Once the IRF and all integration requirements are formally approved by both the RAS Team and the System Owner, the RAS Team will notify the Partner System Team by email that the integration has advanced to the Design phase.



Design Phase Summary & Outputs



Summary

- During the Design phase, the Partner System and RAS Team will flesh out the technical design elements from the IRF to execute on the integration plan outlined in the Planning phase.
- Design principles that are critical to RAS integration are demonstrated in documentation through use cases, user workflows, security requirement considerations, mockups, and architectural diagrams.



Actions and Outputs

1. The Partner System Team will complete the [Preliminary Technical Design Document \(TDD\)](#) and share a copy with the RAS Team for initial review and feedback.
2. After finalization, review, and System Owner sign-off, the Preliminary TDD is submitted for RAS Team review and sign-off.
3. The Partner System Team continues completing or updating/validating the [ISA](#).

Design Phase Playbook

Utilize the RAS
Playbook in tandem
with the RAS SOP



RAS SOP - Design
(Pg. 15)

Step 1: Discuss RAS Integration Design Components

Meet internally to review RAS integration design components and outline the intended integration structure. Please refer to your IRF, as needed, to align consistent language and diagrams for the Design Phase. The elements and questions below are not mandatory to be documented for the RAS SOP, but are meant to help inform the creation of design documents:

- **System Description:** Summarize the system and the primary use cases for integration with RAS.
- **System Architecture:** Include system overviews, sub-system or platform workflows, and user workflows.
 - Are the system architecture diagrams and sequence diagrams available and up to date?
 - If any are missing, who is responsible for creating or updating these diagrams?
 - Have all sub-systems and analysis platforms been included in both system-wide diagrams and called out individually for their particular use cases and workflows?
- **Use Cases:** Clearly describe use cases such as login, analysis, passport management, session handling, and logout.
 - Are all relevant use cases, such as login, analysis, passport management, session management, and logout, clearly documented for each platform or sub-system?
 - If not, who will ensure each platform or sub-system has comprehensive use case documentation and sequence diagrams?
 - Are OAuth code samples provided relevant, with supporting workflows and diagrams to explain the integration?



Design Phase Playbook



Step 1: Discuss RAS Integration Design Components (Continued)

- **Use Cases (Continued):** Clearly describe use cases such as login, analysis, passport management, session handling, and logout.
 - Have user flows and diagrams been developed, and is the reasoning behind each user flow and design choice clearly articulated?
 - Are the practices for passport management, data access, and any anticipated future requirements (such as DRS servers or external clearinghouses) thoroughly explained?
 - Is there a clear explanation of user actions after authentication and authorization, especially regarding data storage, caching, and access controls?
- **User Experience Mockups:** Provide UI/UX mockups that represent typical user workflows and interactions.

Step 2: Convene with RAS Team on Design Details (If Needed)

If additional questions or technical clarification is needed, the Partner System Team schedules a meeting with the RAS Team (NIHRASTeam@mail.nih.gov) to discuss how integration elements from the Planning phase will be implemented in the integration design.

Step 3: Draft the Preliminary Technical Design Document (TDD)

The Partner System Team drafts the [Preliminary TDD](#), integrating outputs from internal discussions and the RAS Team (if applicable). All critical design elements should be captured.

- Utilize the NCPI ACC's [Preliminary TDD tips and tricks template](#) to help with completeness and alignment with best practices.
- (Past Process) Partner Systems can share the Preliminary TDD with the NCPI ACC for an informal RAS review to gain feedback based on prior RAS integrations and help align the TDD with RAS expectations.



Design Phase Playbook



Step 4: Submit Preliminary TDD for RAS Review

Submit the Preliminary TDD, signed off by the System Owner, with NCPI ACC copied, via the designated [Box](#) folder to the RAS Team.

- Email the RAS Team (NIHRASTeam@mail.nih.gov) following the [communication plan](#) guidelines with the document [Box](#) link, confirmation of System Owner approval, and request the expected review turnaround time (typically 1–3 weeks).

Step 5: Understand and Address RAS Review of Preliminary TDD

Due to this is being 'Preliminary' TDD, most feedback from RAS' review can be addressed in the 'Final TDD' in the [Development Phase](#). For the Final TDD, the RAS Team expects all comments to be addressed directly in the document and cleaned up in entirety. This dual draft process allows the integration to move forward more efficiently.

- Please note, some feedback from RAS is required in the Preliminary to ensure that the design makes sense and is compliant. Please address comments and comment the mitigation/acknowledgment for each RAS comment.

Step 5: Confirm RAS Team Sign-Off and Transition to Development

Upon sign-off, the RAS Team confirms that the Preliminary TDD supports best practice design principles and is in adherence to security requirements. Once the RAS Team provides sign-off, the Partner System Team can move into the Development phase.



Development Phase Summary & Outputs



Summary

- Using the RAS Staging environment, the Partner System and RAS Team will construct the integration design.
 - As Partner Systems utilize the RAS Staging environment for the first time, there may be unexpected bugs or edge cases that need to be resolved during the Development phase.
 - Development phase timelines vary based on estimated level of effort evaluated in the Planning Phase, based on the specific requirements of the Partner System.
 - During this phase, it is necessary to have regularly scheduled touchpoints between the Partner System and RAS Teams, which will last through Production release.



Actions and Outputs

1. Partner System Teams receive RAS Staging credentials and test accounts, iron out technical roadblocks, and map test cases in preparation for entering Testing.
2. If there are updates made during development due to technical roadblocks, documentation completed in previous phases (IRF, TDD) must be updated.
3. The Partner System Team will complete the [Test Provisioning Form](#)/[Test Plan](#) and share a copy with the RAS Team for review and feedback. After finalization and with System Owner sign-off, the Partner System Team submits the Test Form and Test Plan to the RAS Team for review and sign-off.
4. Partner System Team continues renewal or development of the [ISA](#).



Development Phase Playbook

Utilize the RAS
Playbook in tandem
with the RAS SOP



RAS SOP - Dev
(Pg. 16)

Step 1: Establish Recurring RAS Integration Touchpoints

The RAS Team will schedule recurring touchpoints, which will extend through the remainder of the integration process.

- Recurring touchpoints provide a forum to address technical issues, architecture questions, and open items, ensuring efficient RAS Integration. If neither the RAS nor Partner System has issues to discuss, the RAS Team may cancel that week's meeting.
- If the Partner System experiences a roadblock between meetings or do not see reoccurring meetings and need to meet with RAS, Partner Systems should reach out to the RAS Team via email (NIHRASTeam@mail.nih.gov)

Step 2: Check on ISA renewal or development status

The Partner System Team should double check on ISA renewal or development status to ensure it is on track for completion by the end of Testing.

Step 3: Finalize and Submit the Technical Design Document (Final TDD)

The Partner System Team reviews all RAS Team feedback, makes any required updates, and completes the Final TDD referencing RAS' feedback from the Design Phase.

- Partner System Team completes the Final TDD by addressing remaining feedback from the RAS Team and uploads it to [Box](#) with System Owner approval.
 - Updates from the Preliminary TDD to Final TDD should only include minor details and document cleanup.
- (Past Process) Partner Systems can share a copy of the Final TDD with the NCPI ACC for an informal RAS review to gain feedback based on prior RAS integrations and help tailor the Final TDD to RAS expectations.



Development Phase Playbook



Step 4: Discuss RAS Integration Development Components

Meet internally as needed, to review RAS integration development components. Refer back to your IRF and TDD as needed, to align consistent language for the Development Phase. The elements and questions below are not mandatory to be documented for the RAS SOP, but are meant to help inform the creation of design documents:

- Purpose of test plan: Summarize the scope of testing for the partner system's integration with RAS (e.g., authentication, authorization, etc.)
- Scope of testing: Who will be performing the testing, What subsystems and authentication/authorization workflows will be tested?
- What RAS Login sources (IdPs) will be tested?
- What test accounts will be needed to appropriately test? (e.g., RAS Accounts, Login.gov Accounts, ID.me Accounts, etc.)
- What are all the types of users that will be using RAS to authenticate/authorize (e.g., first-time users, users that have already utilized RAS' account linking capability, users that haven't linked their accounts, admin users, users who shouldn't have access, etc.)
- If the Partner System has multiple sub-systems/platforms (e.g., Gen3, Velsera, etc.), all sub-system / platform workflows captured?
- What warrants a Pass for each Test Case, and what warrants a Fail for each Test Case?

Step 5: Develop the Test Plan

The Partner System Team drafts the [Test Plan](#), integrating outputs from internal discussions and the RAS Team (if applicable).

- Utilize the [Test Plan NCPI ACC Tips and Tricks](#) template to help with completeness and alignment with best practices.
- (Past Process) Partner Systems can share the Test Plan with the NCPI ACC for an informal RAS review to gain feedback based on prior RAS Integrations and help tailor the Test Plan to RAS expectations.



Development Phase Playbook



Step 6: Submit the Test Plan

Submit the Test Plan, signed off by the System Owner, with NCPI ACC copied, via the designated [Box](#) folder to the RAS Team.

- Email the RAS Team (NIHRASTeam@mail.nih.gov) following the [communication plan](#) guidelines with the document [Box](#) link, confirmation of System Owner approval, and request the expected review turnaround time (typically 1–3 weeks).
- Review and address RAS Team feedback as applicable before proceeding.

Step 7: Complete Test Account Provisioning Form

The Partner System Team fills out the [Test Account Provisioning Form](#), integrating outputs from internal discussions and the RAS Team (if applicable).

- Leverage and reference the Test Plan and [Test Account Provisioning Form NCPI ACC Tips and Tricks](#) template to understand which test accounts are needed to provision and populate the form accordingly.
- Only include test accounts that need to be set up by the RAS Team. These are accounts where the Partner System receives credentials directly from the RAS Team after setup (e.g., RAS accounts, eRA Commons accounts).
 - If dbGaP authorizations are required, request these within a RAS account and the RAS Team will collaborate with the dbGaP team to assign these permissions accordingly.
- Other accounts can be provisioned by the Partner System Team directly from the source (e.g., [Login.gov](#), [ID.me](#) accounts).
- (Past Process) Partner Systems can share the Test Account Provisioning Form with the NCPI ACC for an informal RAS review to gain feedback based on prior RAS integrations and help tailor the Test Account Provisioning Form to RAS expectations.



Development Phase Playbook



Step 8: Submit the Test Account Provisioning Form

Submit the Test Account Provisioning Form, signed off by the System Owner, with NCPI ACC copied if still operational, via the designated Box folder to the RAS Team.

- Email the RAS Team (NIHRASTeam@mail.nih.gov) following the [communication plan](#) guidelines with the document Box link, confirmation of System Owner approval, and request the expected review and provisioning of test accounts (typically 1–3 weeks).
- Review and address RAS Team feedback as applicable before proceeding.

Step 9: Request Staging Environment Connection

Following the retrieval of test accounts, the Partner System Team submits a Client ID/Client Secret request via GitHub to connect their staging environment to the RAS staging environment per the [communication plan](#) guidelines.

Step 10: Obtain Approval to Proceed to Testing

Upon securing sign-off from the RAS Team and System Owner on all development documentation via email (Test Provisioning Form and Test Plan) per the [communication plan](#) guidelines, the Partner System Team advances to the Testing Phase.



Testing Phase Summary & Outputs



Summary

- The Partner System development teams and RAS Team ensure end-to-end testing is completed before production release.
- The Tabletop Exercise (TTX) is conducted by the Partner System and RAS Teams to practice incident response under simulated conditions.



Actions and Outputs

1. The Partner System Team develops the [TTX Scenario](#) in preparation for the TTX.
2. After the TTX, the Partner System Team provides audit logs via email to the RAS Team for review and sign-off.
3. If bugs/issues arise during testing, they are recorded using the [TTX Action Report](#) (located at end of document) and tracked via [GitHub](#) tickets until resolved.
4. Testing concludes with written sign-off from both the Partner System and RAS Teams.
5. After completion of the TTX, execution of items on the [Testing](#) and [Integration Checklists](#) by Partner System Team, and a completed and signed [ISA](#) from both parties, the integration proceeds to Deployment.



Testing Phase Playbook

Utilize the RAS Playbook in tandem with the RAS SOP



RAS SOP - Testing (Pg. 18)

Step 1: Prepare the Tabletop Exercise (TTX) Scenario

The Partner System Team drafts the [Tabletop Exercise \(TTX\) Scenario](#), in preparation for the TTX, which is designed to practice incident response under simulated conditions.

- Reference [TTX Scenario NCPI ACC Tips and Tricks](#) template to understand positive and negative scenarios for testing both Authentication and Authorization with RAS.
- TTX scenarios are designed to comprehensively test the Partner System's use cases for RAS integration, including both positive and negative scenarios in a safe test-driven environment.
- TTX Scenarios may verify the user authentication, IAL level, consent, normal OIDC flows, user authorizations, OIDC claims, and RAS compliance requirements, including all passport-based activities, in a safe, test-driven environment.
- (Past Process) Partner Systems should share a copy of the TTX Scenario with the NCPI ACC for an informal RAS review. This can provide valuable feedback based on prior RAS integration experience and help tailor to RAS expectations.

Step 2: Submit the Tabletop Exercise (TTX) Scenario

Once the TTX Scenario has been prepared and approved by the Partner System team, upload it to [Box](#) for RAS Team review and approval before scheduling the TTX.

- Email the RAS Team (NIHRASTeam@mail.nih.gov) following the [communication](#) guidelines with the document [Box](#) link, confirmation of System Owner approval, and request the expected review turnaround time (typically 1–3 weeks).
- Address RAS Team feedback as applicable.



Testing Phase Playbook



Step 3: Schedule Tabletop Exercise (TTX) Session

The RAS Team coordinates with Partner System development teams to schedule the TTX.

- TTX typically can be completed in 30 minutes. In the case of complex integrations, additional TTX sessions will be scheduled by RAS.
- To have the most effective TTX, make every effort to have both technical and process stakeholders in attendance to participate in the exercise and address any updates needed.

Step 4: Conduct the TTX and Capture Outcomes

The Partner System development teams and RAS Team conduct the TTX using the TTX Scenario submitted previously.

- Auditing and logging requirements will be discussed and expected as the output to the TTX.
- After the TTX, the Partner System Team must compile the finalized [Tabletop Exercise \(TTX\) Scenario](#) along with the Partner System logs and submit them to the RAS Team via email (NIHRASTeam@mail.nih.gov) following the [communication plan](#) guidelines.
- The Partner System Team should report all identified bugs or issues found in the TTX with RAS via [GitHub](#) until resolution and sign-off occurs.



Testing Phase Playbook



Step 5: Complete and Submit the Testing Checklist

The Partner System Team completes the [Testing Checklist](#), confirming coverage of all RAS services and integration points, and uploads it to [Box](#).

- This checklist is intended to ensure that appropriate testing was completed by the Partner System Team for all RAS Services being used in the integration prior to Production.
- Email the RAS Team (NIHRASTeam@mail.nih.gov) following the [communication plan](#) guidelines with the document [Box](#) link, confirmation of System Owner approval, and request the expected review turnaround time (typically 1–3 weeks).
- With this sign-off, the Partner System Team also certifies that necessary testing for negative scenarios was successfully completed, and appropriate responses are obtained from the RAS Team for each test case.

Step 6: Complete and Upload the Integration Checklist

The Partner System Team finalizes the [Integration Checklist](#) and uploads it to [Box](#).

- This checklist is intended to confirm completion of all major integration stage gate activities across both Partner System and RAS Team before Go-Live
- Email the RAS Team (NIHRASTeam@mail.nih.gov) following the [communication plan](#) guidelines with the document [Box](#) link, confirmation of System Owner approval, and request the expected review turnaround time (typically 1–3 weeks).



Testing Phase Playbook



Step 7: Finalize the Interconnection Security Agreement (ISA)

Ensure the [ISA](#) is completed and signed off on by all required stakeholders prior to being approved to the Deployment Phase by RAS.

- **At this point, if the ISA is not completed, finalized, and signed, the RAS Integration cannot continue.**
- To avoid delays, it is advised to begin the ISA [renewal/creation](#) process during the Planning Phase.



Deployment Phase Summary & Outputs



Summary

- Testing transitions to the RAS production environment as the final step in the RAS Integration process before ongoing Operations & Maintenance.
- Deployment into the production environment can begin upon successful completion of testing in pre-production, inclusive of submission of all required documentation.



Actions and Outputs

1. The RAS Team and Partner System Team must work together to coordinate a deployment schedule and conduct end-to-end testing of the production deployment.
2. If defects are detected, they must be reported to the RAS Team and tracked via [GitHub](#) tickets until resolved.
 - Defects within the production environment are considered critical and may be escalated accordingly.
 - The Staging environment closely mimics the production environment but there may be some exceptions, edge cases, and net new features where bugs are discovered after deployment to production.
3. Sign-off by the Partner System Team is required after successful deployment to production.

Deployment Phase Playbook

Utilize the RAS
Playbook in tandem
with the RAS SOP



RAS SOP -
Deployment (Pg. 20)

Step 1: Partner System and RAS Teams determine a feasible deployment schedule based on anticipated production testing needed.

Step 2: The Partner System conducts end-to-end testing in the Production testing environment using production credentials distributed by the RAS Team.

Step 3: In the Production testing environment, the Partner System Team records any issues detected via [GitHub](#) and work towards issue resolution, using recurring touchpoints scheduled during the Development phase, as needed.

Step 4: In preparation for 'Go-Live' Production, Partner System Team should notify their user population of process changes, as necessary, including development of system-specific user guides that address how to report user issues post-production.

Step 5: Once all issues in the Production testing environment are resolved, the RAS Team can initiate 'Go-Live' Production, making RAS production available to all Partner System users.

Step 6: To close Deployment, Partner System Teams email the RAS team to confirm 'Go-Live' Production is complete.



O&M Summary & Outputs



Summary

- During Operations & Maintenance (O&M), Partner System Teams work with the RAS Team to complete ongoing maintenance to the RAS infrastructure.
- Partner System Teams conclude the transition to RAS and increased security posture by sunsetting the use of any legacy Auth tools (e.g., dbGaP telemetry files).
- Partner System Teams confirm that all necessary permissions are maintained in the move to the RAS passport process.



Actions and Outputs

1. Partner System Teams work with the RAS Team on ISA renewal, new feature/enhancement requests, and issue reporting and remediation through [GitHub](#) tickets.
2. Partner System Teams work with the RAS Team to jointly analyze and design the transition process from legacy Auth tools.
3. Partner System Teams monitor interim period where both legacy Auth tool and RAS passports are used, allowing systems to identify and resolve any issues.
4. NCPI IC System Owners formally approve and confirm readiness to decommission legacy Auth tools. RAS Passports are implemented successfully, enabling modernized authorization to NIH CAD.
5. Partner System Teams will provide official notice to the RAS Team indicating the completion of legacy Auth tool sunsetting.



O&M Playbook

Utilize the RAS
Playbook in tandem
with the RAS SOP



RAS SOP - O&M
(Pg. 21)

Step 1: Partner System Teams work to sunset legacy Auth tools.

- Partner System Teams analyze and design the transition process from legacy Auth services. Partner System Teams review the current legacy Auth services, determine which will be handled by RAS, and confirm that all necessary permissions are maintained in the move to the RAS.
- Partner System Teams, IC System Owners, and Authorizing Officials (AOs) should sign off on the transition.
- Once Partner System Teams, IC System Owners, and AOs have signed off that RAS has no gaps, access issues, or security concerns, the Partner System Team should begin the sunset of legacy Auth services.
- The Partner System Teams and their legacy Auth services will halt processes for pulling/uploading of authorizations, including disabling scripts, files, folders, and other file elements for the applicable Partner System.

Step 2: Partner System Teams work with the RAS Team for the following ongoing maintenance activities:

- ISA renewal: RAS Team will be responsible for identifying the need for ISA renewal. A review of the ISA and IRF shall be conducted collaboratively by the RAS Team and the Partner System Team.



Step 2 (Continued): Partner System Teams work with the RAS Team for the following ongoing maintenance activities:

- New feature/enhancement requests: System owners/teams can make these requests on an ongoing basis. The IRF will need to be reviewed and updated; this will trigger a restart of the review process to gauge feasibility. Formal/informal discussions will occur as needed for planned upgrades or enhancements.
 - Report via GitHub: <https://github.com/NIH-Auth-Services/CIT-IAM-RAS/issues/new/choose> (Select 'Feature Request').
 - Additionally, requests can be customized with labels, allowing the RAS Team to better understand the needs of the Partner System.
- Issue reporting via GitHub: the RAS Team allows Partner System Teams to submit support requests via RAS GitHub to help resolve issues such as bugs or unexpected/unannounced downtime for the Partner System.
 - Report via GitHub: <https://github.com/NIH-Auth-Services/CIT-IAM-RAS/issues/new/choose> (Select 'Defect-Bug report' or 'Support request' as applicable).
 - Optional reporting via ServiceNow: IC System Owner may also track issues with the RAS Team by raising a ServiceNow ticket and routing it to "CIT-IAM-RAS-Team" queue.

Document Revision History

Date	Description	Version	Author
12/12/2025	Initial release of NCPI RAS Playbook.	1.0	NCPI ACC
03/03/2026	Incorporated feedback from ODSS Leads.	1.1	NCPI ACC

